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SOFTWARE PROJECT

FoodieNetwork

Cooking Lovers Application

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Chosen technology stack

The chosen technology stack for Application is React and MongoDB for the backend, along with HTML and CSS for the frontend. This combination provides a scalable, flexible, and interactive user experience while ensuring efficient data management.

Backend Technologies

React

React is a free and open-source JavaScript library used for building user interfaces, especially for single-page applications (SPA). It follows a component-based architecture and efficiently updates the UI using a virtual DOM.

Key Features:

Component-Based: UI is divided into reusable components.

Virtual DOM: Optimized rendering improves performance.

State Management: Uses hooks like useState and useReducer.

One-Way Data Binding: Ensures predictable data flow.

React Router: Handles client-side routing in SPAs.

MongoDB

MongoDB is a NoSQL database that stores data in a flexible, JSON-like document format. It provides scalability, high availability, and ease of integration with Node.js and Express.js.

Why MongoDB Instead of Firebase? Unlike Firebase, which is a managed Backend-as-a-Service (BaaS), MongoDB provides greater flexibility for complex queries, on-premise deployment, and indexing, making it ideal for a dynamic application like FoodieNetwork.

Key Features:

Document-Based: Uses JSON-like data storage (BSON format).

Schema Flexibility: Allows dynamic fields and embedded documents.

High Performance: Supports indexing, aggregation, and distributed processing.

Scalability: Horizontal scaling via sharding.

Powerful Querying: Supports complex queries and real-time analytics.

Authentication & Security: Uses role-based access control (RBAC) and encryption.

MongoDB will be integrated using Mongoose, an ODM (Object Data Modeling) library, which simplifies data modeling and validation.

Express.js

Express.js is a minimal and flexible Node.js web framework that provides robust API functionalities. It is used to build the backend server and manage routing.

Key Features:

Middleware Support: Enables custom request handling.

Routing System: Defines API endpoints efficiently.

Lightweight and Fast: Optimized for handling HTTP requests.

Integration with MongoDB: Seamlessly connects with Mongoose.

Node.js

Node.js is a JavaScript runtime that allows running JavaScript code outside the browser, making it ideal for backend development.

Key Features:

Asynchronous & Event-Driven: Handles multiple requests efficiently.

Fast Execution: Uses the V8 engine for quick processing.

NPM Support: Provides access to a vast package ecosystem.

Spoonacular API

The Spoonacular API is a comprehensive food and recipe database that provides access to a vast collection of recipes, ingredient information, nutrition analysis, and meal planning features.

Key Features:

Recipe Search & Recommendations: Users can search for recipes based on ingredients or dietary preferences.

Nutritional Analysis: Provides detailed nutritional breakdowns of recipes and ingredients.

Meal Planning: Users can generate meal plans based on dietary needs and calorie goals.

Ingredient Data: Offers extensive ingredient details, including substitutions and cost analysis.

Spoonacular API will be used in FoodieNetwork to enhance recipe recommendations, enable meal planning features, and provide users with nutritional insights.

Frontend Technologies

HTML

Hypertext Markup Language (HTML) is used to define the structure and layout of web pages. It allows embedding multimedia, forms, tables, and navigation elements.

CSS

Cascading Style Sheets (CSS) manage the styling of the application, ensuring a modern and responsive design

Application architecture

The FoodieNetwork architecture consists of a frontend that interacts with users and a backend that manages data processing and user authentication.

Frontend

Built with React, HTML, and CSS for a responsive UI.

Displays recipes, community interactions (likes, comments, shares), and user profiles.

Backend

Uses MongoDB for data storage.

Implements Express.js for API handling.

Handles user authentication, post management, and interaction processing.

Data Flow

Users interact with the frontend (React) to browse recipes, post content, and engage with others.

Actions (retrieving and saving recipes) trigger API calls to the backend.

The backend processes requests, stores/retrieves data from MongoDB, fetches recipe details from Spoonacular API, and updates the UI accordingly.

Description of Application Functionality

FoodieNetwork includes features to enhance user engagement and simplify recipe sharing:

User Authentication: Secure login and registration.

Home Feed: Displays recipes

Search Functionality: Allows users to search for recipes and profiles.

Recipe Creating and Saving: Allows to create and save recipes

Description of User Interface

Color Theme: Orange (associated with freshness, warmth, and food culture).

Design: A modern, minimalistic UI with interactive buttons/icons.

Pages Included:

Login Page: Allows users to access their accounts.

Sign-Up Page: Enables new users to register.

Welcome page: Displays a menu of pages and redirects to explore page

Explore Page: Search bar with a variety of recipes

Popular Page: Displays popular recipes

Home Page: Allows to post personalized recipes and save other recipes.

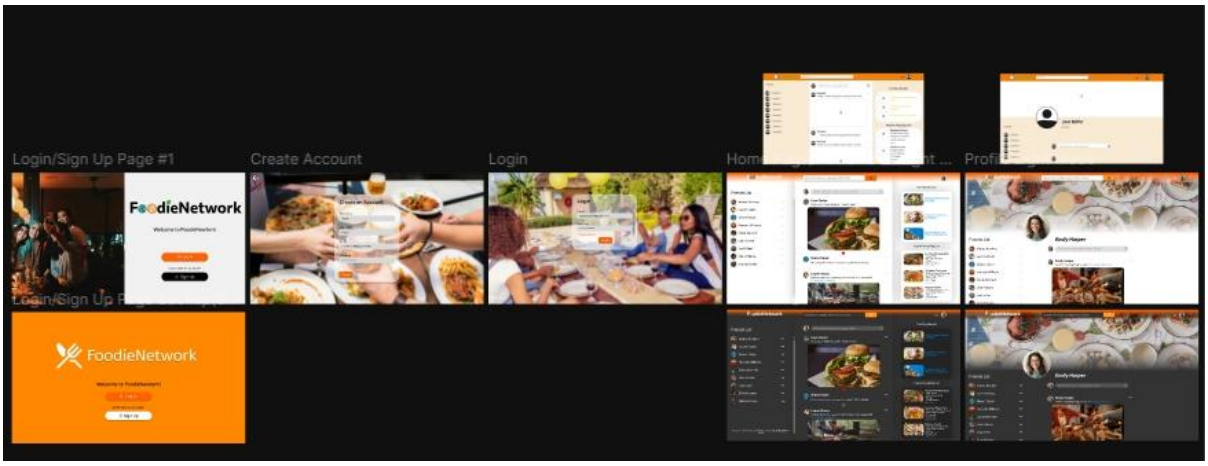

Fig1: Figma UI design

Summary

FoodieNetwork is a social media platform for food lovers to share, discover, and interact with recipes. The tech stack includes React, MongoDB, Express.js, Spoonacular API, Node.js, HTML, CSS, and JavaScript for a scalable, efficient, and user-friendly experience.

Future updates will include mobile compatibility, AI-powered recipe recommendations, and enhanced community engagement tools.

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